GERIATRIC AI DEVICE FOR REMOTE HEALTH MONITORING SYSTEM

A PROJECT REPORT

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ABSTRACT

The rapid increase in the global aging population has intensified the demand for innovative Biomedical and IoT-driven safety systems that support independent living. By 2050, the number of elderly individuals worldwide is expected to reach 1.5 billion, placing enormous pressure on healthcare infrastructure and family caregivers alike. In the domain of remote healthcare, ensuring the safety of elders through continuous vitals monitoring and emergency detection is a critical challenge. Existing methods often rely on high-cost hospital equipment or basic trackers that lack the intelligence to distinguish between normal activities and life-threatening falls. Furthermore, current cloud-dependent solutions often suffer from latency issues during network instability, making real-time emergency response unreliable.

Our proposed solution, Elder-Sense AI, is a low-cost, smart wearable device integrated with an Edge-IoT architecture. Utilizing an ESP32 microcontroller, the device processes sensor data locally via a custom Edge AI algorithm for instantaneous fall detection and heart rate/SpO_2 monitoring. The MAX30100 pulse oximeter sensor continuously measures heart rate and blood oxygen saturation, while the MPU6050 accelerometer and gyroscope module captures body movements to identify sudden falls or prolonged inactivity. This shift from cloud-first to edge-first processing ensures rapid local alerts and reliable remote notifications to caregivers, even in areas with poor network connectivity. Upon

detecting an abnormal health event or a fall, the system simultaneously triggers a local buzzer alarm, dispatches an emergency SMS to the designated caregiver via a GSM module (SIM800L), and uploads the real-time health data to the Thing Speak cloud platform for remote visualization. A dedicated panic button also allows the elderly user to manually raise an alert at any time. The device operates on a rechargeable Li-Po battery and features an LCD display, making it portable, self-contained, and easy to use without any technical knowledge.

Results demonstrate a high-accuracy, low-latency system that is affordable for domestic use and suitable for deployment in both home and institutional care environments. Future enhancements include predictive health analytics powered by machine learning models such as LSTM networks, integration of additional biosensors, and power-saving optimizations for long-term wearability. Clinical validation in collaboration with medical institutions is also planned to ensure real-world reliability and safety compliance.

Keywords: *Geriatric Health Monitoring, Internet of Things (IoT), Edge Computing, Fall Detection, GSM Emergency Alert, Thing Speak, Wearable Device, Biomedical System, Remote Patient Monitoring.*

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LIST OF ABBREVATIONS

S.No	Abbreviation	Expansion
1	**AI**	Artificial Intelligence
2	**IoT**	Internet of Things
3	**SpO_2**	Peripheral Capillary Oxygen Saturation (Blood Oxygen Level)
4	**ESP32**	Epressif Systems Processor 32 Bit (Microcontroller used)
5	**GSM**	Global System for Mobile Communications
6	**SMS**	Short Message Service
7	**LCD**	Liquid Crystal Display
8	**API**	Application Programming Interface
9	**HR**	Heart Rate

CHAPTER 1

INTRODUCTION

Ageing is a natural part of life, but for millions of elderly people living alone, it comes with serious risks that often go unnoticed until it is too late. A sudden fall in the middle of the night, a drop in blood oxygen levels, or a cardiac irregularity — these are situations where a few minutes of delay can change everything. As the global elderly population is projected to double to 1.5 billion by 2050, the gap between the care that seniors need and the care they actually receive is growing wider every day. Most existing solutions fall short in one way or another. Hospital-grade monitoring systems are expensive and impractical for home use. Basic wearable trackers available in the market are too simplistic — they record data but cannot intelligently detect emergencies or send real-time alerts. Cloud-dependent systems, on the other hand, struggle with response delays whenever the network is unstable, which is precisely when reliable alerts matter the most. There is clearly a need for something smarter, more affordable, and built with the elderly user in mind.

This is what motivated us to develop Elder-Sense AI, a compact and low-cost IoT-based health monitoring device designed specifically for geriatric care. The system continuously tracks heart rate and SpO_2 levels using a MAX30100 sensor, detects falls using an MPU6050 accelerometer and gyroscope, and processes all of this data locally on an ESP32 microcontroller. By handling critical decisions at the edge rather than depending on the cloud, the device guarantees near-instant emergency alerts through a GSM module and a local buzzer, while also syncing health data to the Thing Speak cloud platform for caregiver access from anywhere.

The goal of this project is not just to build a working prototype, but to create something that could realistically be used by a real family, in a real home, without any technical expertise. The sections that follow detail the system architecture, component selection, working principle, and the results achieved so far.

1.1 SCOPE OF THE PROJECT

This project covers the design and development of a low-cost IoT Device for real-time elderly health monitoring, fall detection, and emergency alerting.

1.2 OBJECTIVE OF THE PROJECT

The main objective is to build an affordable and reliable health monitoring device that ensures elderly safety through continuous tracking and instant emergency response. The specific objectives are:

- To continuously monitor heart rate and SpO_2 levels using the MAX30100 sensor in real time.
- To detect falls instantly using the MPU6050 accelerometer and gyroscope and trigger local buzzer alerts.
- To send emergency SMS notifications to caregivers via the GSM module upon detecting critical events.
- To upload health data to the Thing Speak cloud platform for remote monitoring and visualisation
- To design a compact, battery-powered device that requires no technical knowledge to operate.

1.3 REPORT SUMMARY

Existing literature highlights a growing demand for smart elderly care solutions but reveals major gaps in affordability, real-time responsiveness, and ease of use. Studies on AI in healthcare show strong potential but note barriers like infrastructure and digital exclusion among older adults. Research on virtual assistants and IoT devices confirms that passive, automated monitoring works better for elderly users than interaction-dependent systems. Works on robotics and AI-based care emphasise the need for privacy-conscious, human-supportive designs rather than replacement technologies. These findings collectively guided our decision to build Elder-Sense AI as a simple, edge-driven, alert-based system.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

A detailed literature survey was conducted to understand the current landscape of IoT-based health monitoring, AI applications in elderly care, and the challenges associated with deploying such systems in real-world environments. The following papers were reviewed to build a strong theoretical foundation for this project and to identify the gaps that Elder-Sense AI aims to address.

2.1.1 - Challenges and Opportunities of Artificial Intelligence in African Health Space

This study evaluates the current roles of AI in African healthcare, identifies key opportunities, and proposes a policy simulation framework tailored to the African context. The research draws from existing healthcare data, policy documents, and case studies to assess where AI can make the most meaningful impact.The paper concludes that AI has genuine potential to transform healthcare productivity and resource allocation in resource-constrained environments. However, it also highlights that achieving this requires overcoming significant challenges related to data privacy, lack of infrastructure, and the need for a strong ethical policy framework before large-scale deployment. Relevance to Our Project: This paper reinforced our understanding that AI-based health solutions must be designed with accessibility and affordability as core priorities — not just technical performance. It shaped our decision to keep the Elder-Sense AI system simple, low-cost, and deployable without requiring sophisticated infrastructure, making it suitable even for rural and semi-urban area.

2.1.2 - A Review of Age-Friendly Virtual Assistive Technologies and Their Effect on Daily Living for Carers and Dependent Adults

Marston and Samuels draw from gerontology, human-computer interaction (HCI), and disability studies to examine how virtual assistive technologies are being used to support independent living for elderly and dependent adults. The review looks at commercial products such as Amazon Alexa and evaluates their real-world usability for older users and their caregivers. Virtual assistants can significantly enhance independence for elderly and disabled individuals while also reducing the emotional burden on caregivers. However, the study identifies digital exclusion as a critical and ongoing barrier — particularly for "slow adopters" and elderly users from marginalised groups who struggle to engage with interaction-dependent technologies.Relevance to Our Project: The finding on digital exclusion directly influenced our design philosophy. Unlike voice assistants or app-based trackers that require active user engagement, Elder-Sense AI is designed to work passively — monitoring in the background and only activating during emergencies. This ensures that even elderly users who are completely unfamiliar with modern technology can benefit from the system without any learning curve.

2.1.3 - Artificial Intelligence and Robotics in Elderly Healthcare — Enabling Independence and Quality of Life

This paper focuses on AI and robotics applications specifically aimed at the elderly population, covering areas such as health monitoring, social interaction support, and mobility assistance. It reviews the deployment of social robots, exoskeletons,

and AI diagnostic tools in institutional as well as home-care environments.

AI and robotics are identified as vital tools for maintaining autonomy and quality of life in an ageing global population. However, the paper raises important ethical concerns around privacy, data security, and the risk of diminishing human connection when technology becomes a substitute rather than a supplement to personal care. The authors argue that regulation and ethical design are non-negotiable for responsible deployment.Relevance to Our Project: This paper confirmed that while advanced robotic solutions offer impressive capabilities, they are not practical for typical home use due to high costs and complexity. It guided us towards a minimalist, sensor-driven approach that supports — rather than replaces — the role of human caregivers. The privacy concern also prompted us to process sensitive health data locally on the device rather than continuously streaming it to external servers.

2.1.4 - AI in Indian Healthcare — From Roadmap to Reality

Das et al. explore the current scope and progress of AI adoption in the Indian healthcare system, with a specific focus on diagnostic accuracy, early disease detection, and public health policy reform. The study maps the gap between the theoretical roadmap for AI in Indian healthcare and its actual ground-level implementation.AI is presented as a practical solution to India's well-known shortage of healthcare professionals and its infrastructure gaps in tier-2 and tier-3 cities. The paper argues that achieving full potential requires a unified digital health infrastructure, stronger regulatory frameworks, and widespread awareness. It highlights that IoT-based monitoring tools have a particularly strong role to play in extending healthcare reach to underserved populations.Relevance to Our Project: Being an India-based project, this paper is particularly relevant as it validates the real need for affordable IoT health monitoring solutions in the Indian context. The

shortage of medical professionals highlighted in this study is one of the core problems that Elder-Sense AI directly addresses — by enabling continuous remote monitoring without requiring a doctor or nurse to be physically present at all times.

2.1.5 - Summary of Literature Gaps Identified

Across all four papers reviewed, a consistent set of gaps emerged — the absence of an integrated, real-time, affordable, and user-passive monitoring solution for elderly individuals in domestic settings. While existing research acknowledges the need for such systems, none of the reviewed solutions fully combine edge-based processing, multi-parameter health sensing, instant GSM alerts, and cloud visualisation into a single compact and low-cost device. Elder-Sense AI is designed specifically to close this gap.

CHAPTER 3

EXSITING SYSTEM

Before developing Elder-Sense AI, a thorough analysis of existing elderly health monitoring methods and systems was conducted. Understanding their limitations was essential in identifying what the proposed system needed to do differently. The following sections outline the key existing approaches and the problems associated with each.

3.1 Manual Checkups and Periodic Clinic Visits

The most widely followed method for elderly health monitoring in India and many parts of the world is still the traditional approach — scheduled visits to hospitals or clinics, and periodic manual checkups at home by a nurse or family member. While this method has been the backbone of elder care for decades, it is fundamentally reactive. Health is only assessed at the time of the visit, meaning that any critical event — a fall, a sudden drop in oxygen levels, or a cardiac irregularity — that occurs between appointments goes completely undetected until it is potentially too late. For elderly individuals living alone, this gap in monitoring can be life-threatening. The system has no ability to provide real-time information, no alert mechanism, and no continuity of care between visits.

3.2 Basic Wearables and Local Data Recording Systems

Some commercially available wearable devices such as basic fitness bands and pulse oximeters are used in households to monitor heart rate and SpO_2 levels. While these devices can measure and display readings, they store data only locally on the device itself and have no mechanism to transmit health information to a

caregiver or cloud platform. The data is passive — it sits on the device and requires the user to actively check it. For an elderly person who may not regularly look at readings or may not understand what abnormal values mean, these devices offer very limited practical safety value. There is no alert system, no remote access, and no automated response when something goes wrong.

3.3 Conventional Fall Detection Systems

Dedicated fall detection systems do exist in the market, typically using simple threshold-based accelerometer algorithms or camera-based surveillance. However, these systems are known for producing a high rate of false alarms — triggering emergency alerts for normal rapid movements such as sitting down quickly, bending over, or stumbling without actually falling. This not only causes unnecessary panic for caregivers but also leads to alert fatigue, where repeated false alarms cause people to start ignoring or dismissing notifications altogether. Camera-based systems additionally raise serious privacy concerns, as elderly individuals may be uncomfortable with constant video surveillance inside their own homes. These systems also work in isolation — they detect falls but do not simultaneously monitor vital health parameters.

3.4 Cloud-Dependent Health Monitoring Platforms

More advanced existing systems, often used in hospital or institutional settings, rely entirely on continuous cloud connectivity to process sensor data and trigger alerts. While these systems offer remote monitoring capabilities, they suffer significantly during periods of network instability or internet outages. In a health emergency, a few seconds of connectivity lag can mean the difference between a timely response and a missed one. These platforms also tend to be expensive,

requiring recurring subscription fees and dedicated hardware that is far beyond the budget of most Indian households. Furthermore, they are designed primarily for professional healthcare environments and are not user-friendly enough for routine domestic use by elderly individuals or non-technical caregivers.

3.5 Lack of Multi-Parameter Integration

Perhaps the most significant shortcoming of existing systems is that they tend to address only one problem at a time. A fall detector only detects falls. A pulse oximeter only measures SpO_2. A heart rate monitor only tracks heart rate. None of the commonly available or affordable solutions combine all of these into a single unified system that monitors multiple parameters simultaneously and responds intelligently across all of them. In real life, health emergencies rarely occur in isolation — a fall may be accompanied by a drop in oxygen levels, or an irregular heart rate may be the precursor to a collapse. A truly effective monitoring system needs to see the full picture, not just one piece of it.

3.6 Threshold-Based Fall Detection Systems

Fall detection represents a crucial component of elderly safety monitoring, and numerous systems have been developed to identify fall events using motion sensors. Traditional fall detection devices typically utilize accelerometers to measure sudden changes in motion intensity. When measured acceleration exceeds predefined thresholds, the system classifies the event as a fall and triggers an alarm. Although such systems are simple to implement and computationally efficient, they suffer from significant accuracy limitations.

Human movement is inherently complex, and many routine activities such as sitting abruptly or bending quickly may produce acceleration patterns similar to falls.

Consequently, threshold-based systems frequently generate false alarms, reducing user trust and system reliability. Conversely, some fall events may occur with lower impact magnitude and remain undetected. Additionally, many existing fall detection devices operate independently of physiological monitoring, providing limited information regarding the user's health condition following a fall. The inability to integrate motion analysis with physiological assessment restricts their capability to provide comprehensive emergency evaluation.

3.7 Summary of Existing System Limitations

In summary, existing methods of elderly health monitoring suffer from five core problems — they are reactive rather than proactive, they lack real-time remote alerting, they produce unreliable fall detection, they depend heavily on stable cloud connectivity, and they fail to integrate multiple health parameters into one coherent system. These limitations collectively leave a large and dangerous gap in elderly safety that no single existing solution adequately addresses. Elder-Sense AI is designed specifically to fill each one of these gaps in a way that is affordable, compact, and practical for everyday home use.

CHAPTER 4

PROPOSED WORK

4.1 Conceptual Design Philosophy

The proposed Elder-Sense AI system is designed as an autonomous, intelligent, and energy-efficient healthcare monitoring platform that integrates biomedical sensing, motion analysis, and embedded edge intelligence within a compact wearable architecture. The fundamental design philosophy of the system is centered on localized decision-making combined with selective remote supervision. Rather than relying on centralized infrastructure for data analysis, the proposed solution performs real-time processing directly within the embedded device, thereby ensuring immediate emergency detection and reliable operation independent of network conditions. The system is conceived to function as a continuous safety companion capable of monitoring physiological state, identifying hazardous motion events, and initiating timely alerts without requiring user intervention. This approach directly addresses the limitations of existing monitoring systems by combining affordability, operational autonomy, and comprehensive multimodal monitoring within a unified platform suitable for domestic deployment.

4.2 Edge-Enabled Local Intelligence Architecture

A central innovation of the proposed system is the implementation of edge computing for real-time data analysis and decision generation. The device incorporates an embedded processing unit capable of executing lightweight artificial intelligence algorithms optimized for resource-constrained hardware environments. Sensor data acquired from physiological and motion sensing modules is processed locally through a structured computational pipeline consisting of signal conditioning, feature extraction, pattern evaluation, and event classification. By performing inference

directly at the device level, the system eliminates dependence on remote cloud infrastructure for critical decision-making. This localized processing significantly reduces detection latency and ensures that emergency recognition occurs instantaneously, even in environments with limited or unstable connectivity. Furthermore, edge intelligence enhances privacy by preventing continuous transmission of sensitive physiological data. The localized architecture therefore directly resolves the limitations associated with cloud-centric healthcare monitoring systems while improving responsiveness and operational reliability.

4.3 Multimodal Physiological Monitoring Methodology

The proposed system integrates continuous monitoring of vital physiological parameters as a fundamental component of elderly safety supervision. Optical sensing technology is employed to measure heart rate and blood oxygen saturation through non-invasive photoplethysmographic analysis. The system continuously captures pulse waveforms and oxygenation signals, which are processed locally to derive physiological indicators such as pulse rate variability and oxygen saturation trends. These parameters are evaluated against medically defined thresholds as well as temporal variation patterns to identify abnormal physiological states that may indicate cardiovascular or respiratory distress. By incorporating continuous monitoring rather than periodic measurement, the system enables early detection of deteriorating health conditions. The integration of physiological monitoring within the same platform that performs fall detection ensures comprehensive safety evaluation rather than isolated parameter observation. This multimodal methodology directly addresses the limitation of existing systems that monitor either motion or physiological state independently.

4.4 Intelligent Fall Detection and Activity Classification Strategy

The proposed system employs an advanced fall detection methodology based on multi-parameter motion analysis rather than simple threshold-based evaluation. The motion sensing unit continuously captures acceleration and orientation data, which are processed to derive motion magnitude, posture transition patterns, and impact characteristics. The fall detection algorithm evaluates these parameters collectively, identifying characteristic motion sequences associated with accidental falls. The detection process includes assessment of rapid acceleration changes, abrupt orientation transitions, and subsequent inactivity periods to confirm the occurrence of a fall event. By utilizing multi-stage verification, the system reduces false alarm generation and improves detection reliability compared to conventional threshold-based methods. Additionally, the device continuously classifies routine activities such as walking, sitting, or lying down, thereby establishing contextual awareness that enhances classification accuracy. This intelligent motion analysis strategy resolves the limitations of existing fall detection systems that suffer from poor reliability and high false positive rates.

4.5 Event-Driven Communication and Remote Supervision Framework

To provide caregiver awareness without imposing continuous connectivity requirements, the proposed system implements an event-driven communication architecture. Rather than transmitting raw sensor data continuously, the device communicates only when significant events or abnormal conditions are detected. Upon identification of an emergency, the system generates a structured alert containing event classification, physiological measurements, and time reference, which is transmitted through wireless communication to designated recipients. Periodic health summaries may also be transmitted at predefined intervals for remote supervision. This selective communication strategy minimizes bandwidth usage, reduces power consumption, and

enhances operational reliability during connectivity fluctuations. By avoiding continuous data streaming, the system directly addresses the limitations of existing IoT healthcare solutions that depend heavily on constant network availability.

4.6 Energy-Efficient Embedded Operation Strategy

The proposed system incorporates adaptive power management techniques to enable long-term wearable deployment. The device operates in hierarchical energy modes that regulate sensor sampling frequency, processing intensity, and communication activity according to detected conditions. During normal operation, the system maintains low-power monitoring mode with periodic sampling and minimal processing load. When significant motion or physiological deviation is detected, the device transitions to active analysis mode, temporarily increasing computational intensity to evaluate potential emergencies. Communication modules remain inactive except during alert transmission or scheduled updates. This energy-efficient operational strategy significantly extends battery life and improves practicality for continuous use, thereby addressing the high energy consumption limitation of existing monitoring systems.

4.7 Autonomous Operation and Reliability Enhancement

The Elder-Sense AI system is designed to function autonomously without requiring continuous external support or user intervention. All essential monitoring, analysis, and alert generation processes are executed locally within the device. This autonomy ensures uninterrupted operation even in the absence of network connectivity or caregiver interaction. Reliability is further enhanced through multi-parameter verification of emergency conditions, structured processing cycles that prevent computational conflicts, and redundant alert mechanisms that combine local warnings with remote notifications. The system’s autonomous operational capability directly

addresses the reliability limitations of conventional monitoring systems that depend on external infrastructure for functionality.

4.8 Privacy-Preserving Data Handling Mechanism

To address privacy concerns associated with remote healthcare monitoring, the proposed system adopts a privacy-preserving data handling methodology in which raw physiological data remains within the device. Only processed event information or summarized health metrics are transmitted externally when required. By minimizing data exposure and reducing reliance on centralized storage, the system enhances user confidentiality and reduces vulnerability to data breaches. This localized data handling approach resolves the privacy and security limitations inherent in cloud-dependent monitoring platforms.

4.9 Cost-Effective and Scalable Implementation Approach

Affordability and scalability are key objectives of the proposed solution. The system is constructed using compact embedded hardware and integrated sensing modules that provide high functionality without requiring expensive infrastructure. The simplified hardware architecture and localized processing capability reduce system complexity and maintenance requirements. As a result, the device can be deployed in domestic environments with minimal setup, making continuous monitoring accessible to a wider population. This cost-effective design directly addresses the accessibility limitations of conventional healthcare monitoring technologies.

4.10 Comprehensive Resolution of Identified Limitations

Through the integration of edge intelligence, multimodal sensing, energy-efficient operation, and autonomous decision-making, the proposed Elder-Sense AI system provides a comprehensive solution to the limitations identified in existing

monitoring technologies. Local processing eliminates latency and network dependence, intelligent multi-parameter analysis improves detection accuracy, event-driven communication reduces energy consumption, and integrated physiological and motion monitoring enables holistic safety assessment. The resulting system provides reliable, responsive, and practical healthcare supervision suitable for continuous domestic deployment.

CHAPTER 5

SYSTEM SPECIFICATION

5.1 Component Selection and Design

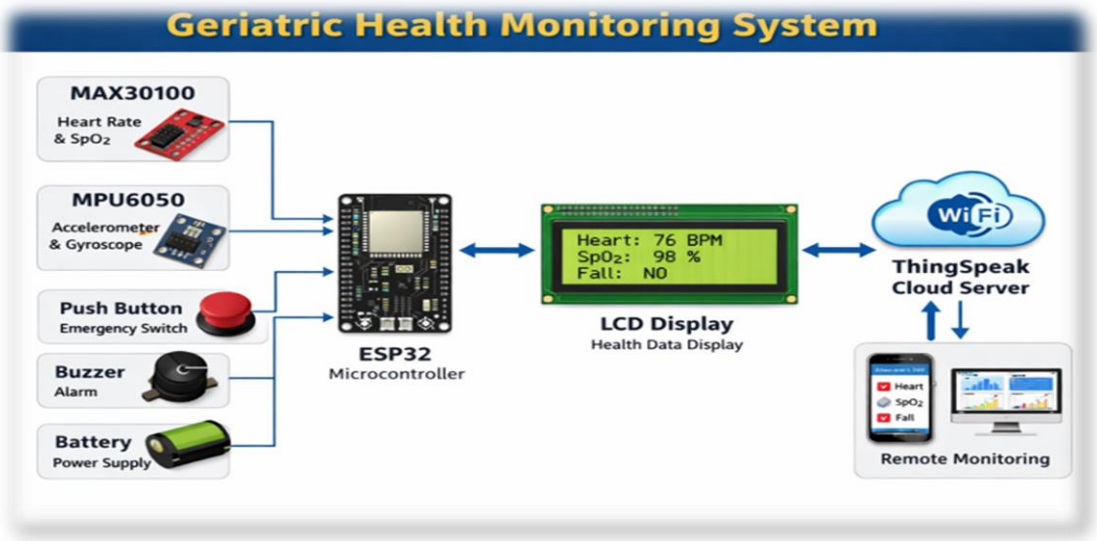


Figure 5.0 - Architecture

The fabrication of the Elder-Sense AI system is guided by the design objectives of low power consumption, compact size, affordability, real-time processing capability, and reliable biomedical sensing performance suitable for continuous domestic health monitoring. The device is constructed around a high-efficiency embedded microcontroller platform featuring integrated wireless communication, sufficient computational capability for edge intelligence processing, and low-power operation modes that support long-term wearable deployment. The microcontroller serves as the central control unit responsible for sensor data acquisition, signal processing, decision-making, and communication management. Its selection is optimized to balance processing capability and energy efficiency, ensuring that real-time analysis can be performed locally without requiring high-power computational resources.

The physiological sensing module is implemented using an optical photoplethysmography sensor capable of simultaneously measuring heart rate and blood oxygen saturation. This sensor is selected due to its non-invasive measurement method, compact footprint, and high sensitivity to microvascular blood flow variations. The module integrates internal signal amplification and optical filtering mechanisms to reduce ambient light interference, thereby improving measurement reliability during continuous monitoring. Its low operating voltage and minimal current consumption make it suitable for wearable healthcare devices where energy efficiency is critical.

Motion sensing is achieved using a multi-axis inertial measurement unit consisting of an accelerometer and gyroscope. The accelerometer measures linear acceleration along orthogonal axes, while the gyroscope captures angular velocity and orientation changes. This combined sensing capability allows accurate characterization of body movement, posture transitions, and impact events necessary for fall detection. The chosen inertial sensor is optimized for low noise performance, high sensitivity to motion dynamics, and minimal power usage, enabling continuous activity monitoring without significantly affecting battery life.

To support real-time alerts and user awareness, the system incorporates local output components including an audible alert mechanism and visual status indicators. These components provide immediate feedback during abnormal events and assist in system diagnostics during operation. Wireless communication functionality is integrated within the embedded controller to enable event-based transmission of alerts and health summaries to caregivers or monitoring platforms. The communication architecture is designed to minimize transmission frequency, thereby reducing power consumption and ensuring reliable operation under varying connectivity conditions.

Power supply is provided through a compact rechargeable energy source supported by voltage regulation circuitry that ensures stable operation of all electronic components. The power subsystem includes energy management features such as

controlled charging, regulated output delivery, and protection against voltage fluctuations. The selection of power components prioritizes operational safety, extended usage duration, and compatibility with low-power embedded hardware.

5.2 Electrical Fabrication and Circuit Integration

The electrical fabrication process involves the systematic integration of sensing modules, processing unit, communication interface, and power management circuitry onto a compact hardware platform. The circuit design is structured to ensure signal integrity, noise minimization, and efficient power distribution. Sensor outputs are connected to the processing unit through dedicated input channels configured for high-resolution data acquisition. Proper grounding and shielding practices are implemented to reduce electromagnetic interference that could affect sensitive physiological measurements.

Signal conditioning pathways are incorporated to stabilize sensor outputs before digital processing. These pathways include filtering stages that suppress high-frequency noise and transient disturbances commonly present in wearable monitoring environments. The circuit layout is optimized to minimize trace length between sensors and processing components, thereby reducing signal attenuation and improving response accuracy.

Power distribution architecture is carefully engineered to maintain stable voltage levels across all modules while preventing excessive energy consumption. The electrical design incorporates low-dropout regulation, controlled current pathways, and protective elements that enhance system reliability. Component placement is arranged to balance thermal distribution and maintain compact device geometry. The electrical integration process ensures that all subsystems operate cohesively within a unified embedded platform capable of continuous real-time monitoring.

5.3 Mechanical Fabrication and Structural Design

The mechanical fabrication of the Elder-Sense AI device focuses on ergonomic wearability, structural durability, and environmental protection of electronic components. The device is enclosed within a lightweight, compact housing designed to be comfortably worn or positioned close to the user without restricting movement. The enclosure is fabricated from non-conductive, biocompatible material that prevents electrical exposure and ensures safe long-term contact with the human body.

The internal layout of the enclosure is designed to secure electronic modules firmly while maintaining appropriate spacing for thermal dissipation. Sensor positioning is carefully considered to maximize contact stability and measurement accuracy. Optical sensing components are aligned to maintain consistent skin contact, while motion sensors are centrally positioned to capture representative body movement patterns. The enclosure incorporates access points for charging and system maintenance without compromising structural integrity.

Shock resistance and vibration tolerance are integrated into the mechanical design to protect internal circuitry from accidental impacts or daily movement stresses. The overall structure balances compactness and accessibility, ensuring that the device remains practical for daily use while maintaining durability under continuous operation conditions.

5.4 Components and Specifications

5.4.1 ESP32 Microcontroller Unit



Figure 5.1 – Microcontroller Unit

The ESP32 serves as the central processing and communication unit of the proposed Elder-Sense AI system, integrating sensing control, signal processing, decision-making, and wireless data transmission within a single embedded platform. It is a dual-core, low-power microcontroller with built-in Wi-Fi and Bluetooth capabilities, making it suitable for real-time IoT healthcare applications. The processor operates at clock frequencies up to 240 MHz, providing sufficient computational capability to execute edge-based algorithms for fall detection and physiological signal processing. The ESP32 includes multiple analog and digital input/output pins, enabling seamless integration with biomedical and motion sensors. It supports deep sleep and power-saving modes, which significantly reduce energy consumption during idle operation and extend battery life. Its onboard memory architecture, consisting of SRAM and flash storage, allows efficient buffering and processing of sensor data streams. The integrated wireless communication module enables reliable data transmission to cloud platforms and remote caregivers without requiring external networking hardware. The ESP32 is selected for this project due to its high performance-to-power ratio, compact architecture, and cost-effective deployment capability.

Key Specifications:

• Processor: Dual-core Tensilica LX6

• Clock Frequency: Up to 240 MHz

• Operating Voltage: 3.0 V – 3.6 V

• Wireless Connectivity: Wi-Fi 802.11 b/g/n, Bluetooth v4.2

• GPIO Pins: Multiple digital and analog interfaces

• Power Modes: Active, Modem Sleep, Deep Sleep

• Memory: Integrated SRAM and external flash support

5.4.2 MPU6050 Motion Sensing Module

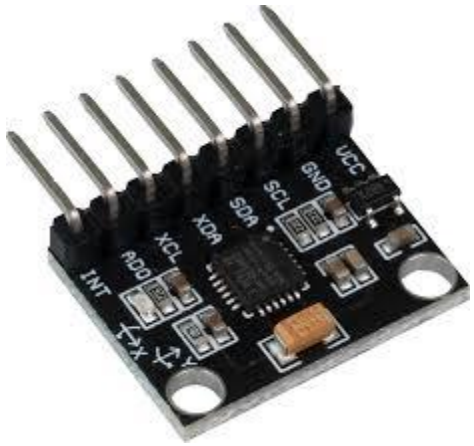


Figure 5.2 – Motion Sensing Module

The MPU6050 is a six-axis inertial measurement unit utilized for motion analysis and fall detection within the monitoring system. It integrates a three-axis accelerometer and a three-axis gyroscope within a compact microelectromechanical system architecture. The accelerometer measures linear acceleration across orthogonal axes, enabling detection of movement intensity and impact characteristics. The gyroscope measures angular velocity, allowing accurate evaluation of orientation changes and posture transitions. Sensor fusion of these measurements enables identification of abnormal motion patterns associated with accidental falls. The module communicates with the microcontroller through a digital interface, providing continuous motion data at configurable sampling rates. Its low power consumption and high sensitivity make it well suited for wearable healthcare monitoring applications requiring continuous activity tracking.

Key Specifications:

• Sensor Type: 3-axis accelerometer + 3-axis gyroscope

• Acceleration Range: ±2 g to ±16 g

• Gyroscope Range: ±250°/s to ±2000°/s

• Communication Interface: I2C

• Operating Voltage: 3.3 V

• Power Consumption: Low-power operation mode available

• Embedded Digital Motion Processor support

5.4.3 MAX30100 Physiological Monitoring Sensor



Figure 5.3 - Physiological Monitoring Sensor

The MAX30100 is a biomedical sensing module designed for non-invasive measurement of heart rate and blood oxygen saturation. It combines optical emitters and a photodetector to perform photoplethysmographic signal acquisition through light absorption characteristics of blood flow. The sensor emits light at specific wavelengths and measures reflected intensity variations caused by pulsatile blood flow, enabling estimation of physiological parameters. The module integrates signal conditioning circuitry, temperature compensation, and ambient light cancellation mechanisms to improve measurement stability. Its compact size and low energy requirement make it suitable for wearable health monitoring devices. Continuous physiological monitoring supports early detection of abnormal cardiovascular or respiratory conditions within the system.

Key Specifications:

• Measurement Parameters: Heart Rate and SpO_2

• Sensing Method: Optical photoplethysmography

• Communication Interface: I2C

• Operating Voltage: 1.8 V (core), 3.3 V (I/O)

• Integrated LED and photodetector

• Low-power biomedical sensing operation

5.4.4 Lithium-Polymer (Li-Po) Battery



Figure 5.4 – Battery

The Li-Po battery functions as the primary power source of the prototype, providing portable and rechargeable energy supply for continuous wearable operation. Lithium-polymer batteries are selected due to their high energy density, lightweight construction, and stable discharge characteristics. The battery supports sustained operation of sensing, processing, and communication components while maintaining compact form factor suitable for wearable integration. Rechargeable capability allows repeated long-term usage without frequent replacement. The battery output is regulated to ensure stable voltage supply to the microcontroller and sensors, thereby preventing performance fluctuations during operation.

Key Specifications:

• Battery Type: Lithium-Polymer Rechargeable

• Nominal Voltage: 3.7 V

• Energy Density: High energy-to-weight ratio

• Form Factor: Compact and lightweight

• Recharge Cycle Capability: Multiple charge-discharge cycles

• Stable discharge characteristics for embedded systems

5.4.5 LCD Display Module



Figure 5.5 – Display Module

The LCD display provides local visual feedback to the user by presenting real-time physiological readings, system status information, and alert notifications. It enables immediate awareness of monitored parameters without requiring external devices. The display is interfaced with the microcontroller through digital communication, allowing efficient data transfer and low power operation. The module is selected to enhance user interaction and system transparency by providing on-device information visualization.

Key Specifications:

• Display Type: Liquid Crystal Display

• Interface: I2C / Parallel interface (depending on module)

• Operating Voltage: 3.3 V – 5 V

• Display Capability: Alphanumeric or graphical output

• Low-power visual output

5.4.6 Software Development Environment – Arduino IDE

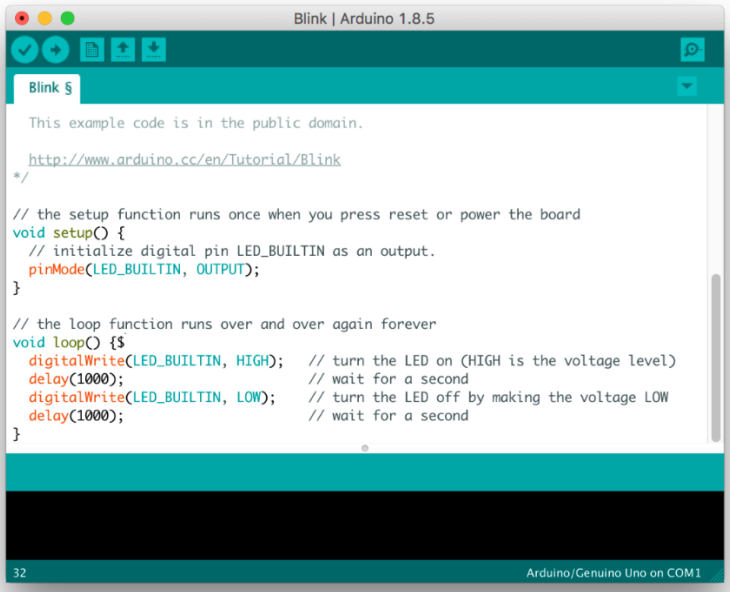


Figure 5.6 - IDE

The Arduino IDE serves as the primary software development platform for firmware implementation, system integration, and embedded algorithm deployment. It provides a simplified programming environment for microcontroller configuration, sensor interfacing, and communication control. The development environment supports compilation, debugging, and serial monitoring, enabling efficient testing and refinement of system functionality. Its extensive library ecosystem facilitates rapid integration of sensor drivers and communication protocols. The platform is selected for its accessibility, cross-platform compatibility, and strong support for embedded IoT development.

Key Features:

• Embedded C/C++ programming support

• Library-based hardware integration

• Serial debugging and monitoring

• Cross-platform development capability

5.4.7 Cloud Data Platform – ThingSpeak

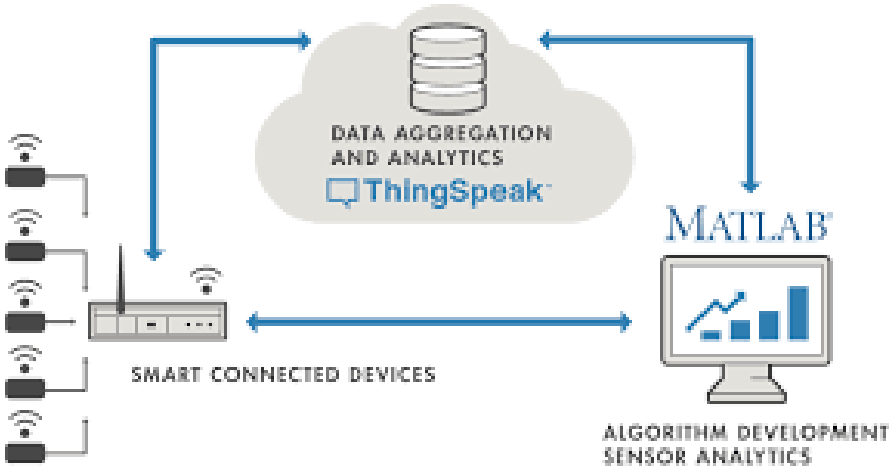


Figure 5.7 – ThingSpeak

ThingSpeak is utilized as the cloud-based data visualization and remote monitoring platform for the proposed system. The ESP32 transmits selected physiological and event data to the cloud using wireless communication, enabling caregivers to monitor user status remotely. The platform supports real-time data visualization, storage, and analysis through web-based dashboards. Event-triggered data transmission reduces bandwidth usage while ensuring that critical health information is accessible for remote supervision. The integration of cloud services enhances system connectivity without imposing continuous data streaming requirements.

Key Features:

• Real-time IoT data visualization

• Secure remote data storage

• Web-based monitoring dashboard

• API support for device communication

• Event-driven data logging capability

5.4.8 Communication Module – GSM Module



Figure 5.8 – GSM Module

The GSM module functions as a cellular communication interface that enables the Elder-Sense AI system to transmit emergency alerts and health status notifications over mobile networks when Wi-Fi connectivity is unavailable or unstable. It provides reliable long-range communication capability by utilizing Global System for Mobile Communications (GSM) infrastructure, ensuring uninterrupted connectivity for remote caregiver notification. The module interfaces with the microcontroller through serial communication and operates using standard AT command protocols for message transmission and network management. Its integration enhances system reliability by introducing a redundant communication pathway, thereby improving emergency responsiveness in real-world deployment environments where internet access may be inconsistent.

Key Features:

• Quad-band GSM/GPRS network support (850/900/1800/1900 MHz)

• SMS-based alert and notification capability

• UART serial communication interface (TX/RX)

• AT command-based control and configuration
• Low idle power consumption with high peak transmission capability
• External antenna support for improved signal reception
• SIM card-based cellular network connectivity

5.5 System Optimization and Integration Strategy

The fabrication process is guided by an optimization strategy that simultaneously addresses performance, energy efficiency, cost, and usability. Component selection prioritizes multifunctional modules that reduce hardware complexity while maintaining functional capability. Local processing architecture eliminates the need for continuous external computation, thereby reducing communication overhead and improving system responsiveness. The integration of sensing, processing, and communication within a single compact platform minimizes system footprint and enhances reliability.

Thermal management considerations are incorporated through component spacing and low-power operation modes, preventing heat accumulation during continuous monitoring. Power optimization techniques ensure that sensing, processing, and communication activities occur only when required, thereby extending operational lifetime. The device is designed to maintain functional stability across varying environmental conditions encountered in domestic settings.

The final fabricated system represents a cohesive integration of biomedical sensing technology, embedded computing capability, and wireless communication within a wearable form factor. The fabrication approach emphasizes simplicity of assembly, scalability of production, and adaptability for future enhancements such as predictive health analytics or advanced sensor integration.

5.6 Software and System Integration

The software upload and integration process establishes the functional intelligence of the Elder-Sense AI device by embedding sensing control logic, signal processing routines, edge decision algorithms, and communication protocols within the microcontroller environment. The firmware is developed using an embedded programming framework that supports real-time sensor interfacing, interrupt-driven processing, and low-power operational modes. During integration, sensor drivers are configured to acquire physiological and motion data at optimized sampling rates that balance measurement accuracy with energy efficiency. Signal conditioning and feature extraction routines are compiled into the firmware to perform noise suppression, peak detection, and motion vector computation directly within the device. The fall detection and physiological anomaly recognition algorithms are implemented as lightweight inference modules that continuously evaluate extracted features against adaptive thresholds and temporal patterns. Communication routines are integrated to enable event-driven wireless transmission of alert packets and periodic health summaries, ensuring that only meaningful data is transmitted to remote recipients. Firmware upload is conducted through a secure serial programming interface, after which system-level calibration parameters, operational thresholds, and alert configurations are initialized. The integrated software architecture operates as a layered control structure in which sensing, processing, decision-making, and communication functions execute in coordinated cycles, thereby enabling autonomous, real-time healthcare monitoring within a compact embedded platform.

5.7 Prototype Assembly

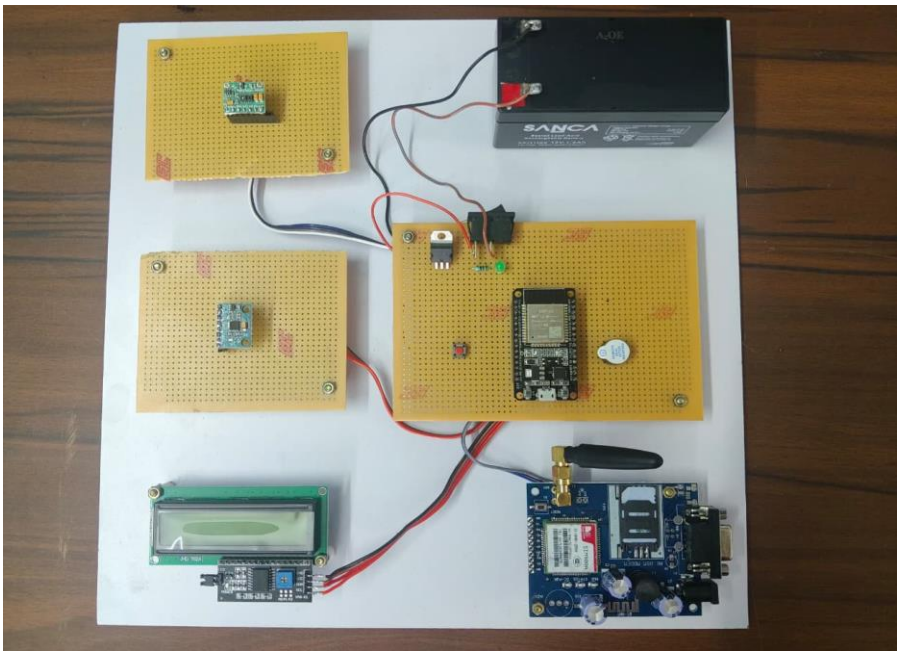


Figure 5.9 – Assembly

The prototype assembly of the Elder-Sense AI system involves the systematic integration of electronic modules, structural components, and power subsystems into a unified wearable monitoring device. The assembly process begins with mounting sensing modules and the embedded processing unit onto a compact circuit platform designed to maintain signal integrity and efficient power distribution. Electrical interconnections are established using secure soldering techniques and insulated wiring pathways to prevent short circuits and minimize electromagnetic interference. Following electrical integration, the assembled circuitry is positioned within a protective enclosure that ensures mechanical stability and safe proximity to the user's body. Sensor placement within the enclosure is carefully aligned to maintain consistent physiological contact and accurate motion capture, while internal spacing is preserved to facilitate heat dissipation and prevent component stress during operation. The power subsystem is incorporated with regulated output pathways to supply stable voltage levels across all modules. External access points for charging and maintenance are

incorporated without compromising structural durability. The final assembled prototype undergoes functional initialization, confirming that sensing, processing, and communication modules operate cohesively within the integrated system architecture. This assembly approach results in a compact, ergonomic, and operationally stable device suitable for continuous domestic deployment.

5.8 Testing and Validation

Testing and validation procedures are conducted to evaluate the operational performance, accuracy, responsiveness, and reliability of the fabricated Elder-Sense AI device under simulated real-world conditions. Functional testing begins with verification of sensor performance through controlled measurement scenarios in which physiological readings and motion patterns are observed across varying user activities. Calibration procedures are applied to ensure that heart rate and oxygen saturation measurements correspond to expected physiological ranges, while motion sensing outputs are validated through structured posture transitions and controlled impact events. The fall detection algorithm is evaluated through repeated experimental trials involving normal daily movements and simulated fall scenarios to assess classification accuracy and false alarm reduction capability. System latency is measured by analyzing the time interval between event occurrence and alert generation, confirming the effectiveness of localized edge processing. Communication reliability is validated by testing event notification transmission across varying connectivity conditions to ensure successful remote alert delivery. Energy performance is assessed through continuous operation testing to evaluate battery consumption patterns and operational lifetime. Validation results confirm that the integrated system maintains stable functionality, rapid response capability, and consistent measurement performance within domestic monitoring environments.

5.9 Safety and Reliability Considerations

The design of the Elder-Sense AI system incorporates multiple safety and reliability mechanisms to ensure dependable long-term operation in real-world healthcare monitoring scenarios. Electrical safety is achieved through insulated circuitry, regulated power distribution, and protective elements that prevent voltage fluctuations or short-circuit conditions. The device enclosure is constructed from non-conductive materials that provide physical protection against environmental exposure while ensuring safe contact with the user. Algorithmic reliability is enhanced through multi-parameter verification strategies that confirm emergency events using both physiological and motion indicators, thereby reducing false alarms and improving detection confidence. Autonomous local processing ensures that emergency detection and alert generation remain functional even during network disruptions, thereby preserving operational continuity. Power management strategies prevent overheating and excessive energy depletion by regulating sensor sampling rates and communication activity. System stability is further supported by structured firmware execution cycles that maintain consistent operation without processing conflicts or data loss. Collectively, these safety and reliability considerations ensure that the device functions as a trustworthy monitoring companion capable of continuous supervision without compromising user well-being or operational integrity.

5.10 Fabrication Outcome and Functional Readiness

Upon completion of electrical and mechanical assembly, the fabricated device functions as a fully integrated Edge-IoT health monitoring unit capable of continuous physiological observation and real-time safety detection. All components operate within a coordinated framework that supports autonomous decision-making, immediate alert generation, and reliable remote communication. The fabrication methodology ensures that the system achieves its intended performance objectives while maintaining

affordability, energy efficiency, and user comfort. Through careful selection of components, optimized circuit integration, and ergonomic structural design, the fabricated Elder-Sense AI device provides a practical and deployable solution for continuous elderly health monitoring in real-world environments.

CHAPTER 6

IMPLEMENTATION AND RESULTS

6.1 Experimental Setup and Evaluation Framework

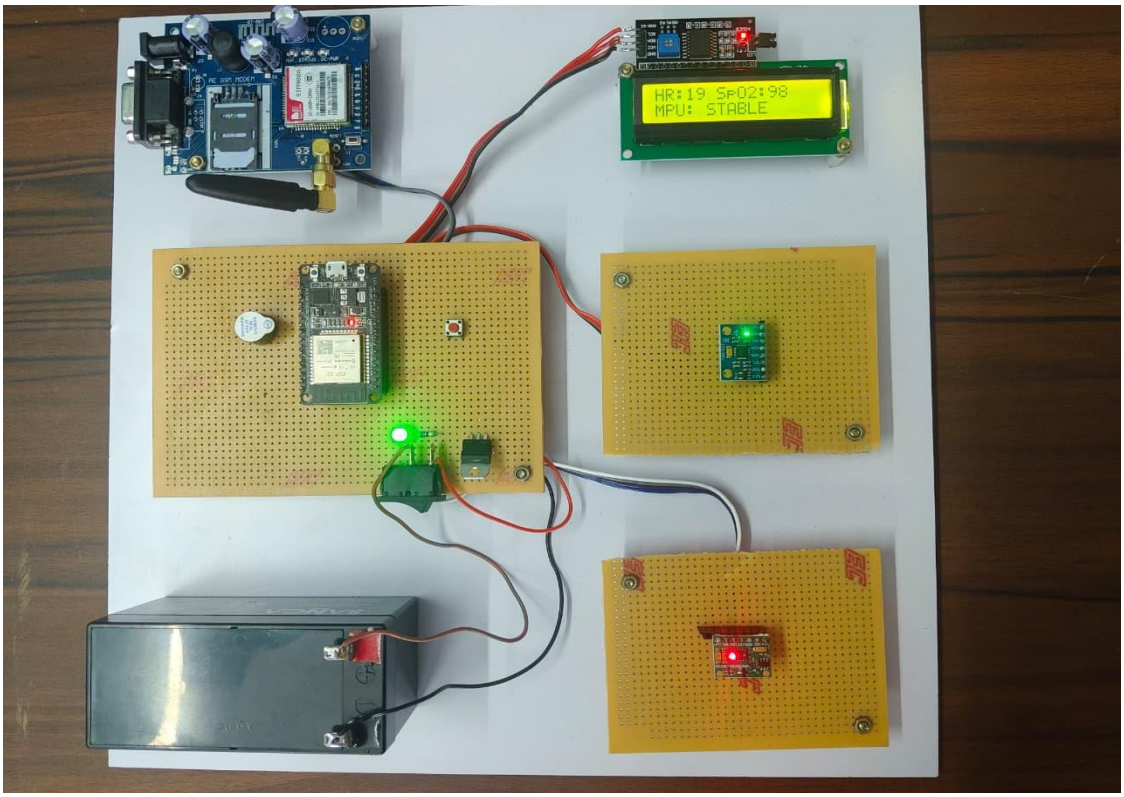


Figure 6.1 – Working Model

The performance of the Elder-Sense AI Edge-IoT health monitoring system was evaluated through a structured experimental framework designed to assess sensing accuracy, event detection reliability, computational latency, communication efficiency, and operational stability under realistic usage conditions. The prototype device was configured to continuously acquire physiological and motion data while executing the embedded edge inference pipeline. Controlled experiments were conducted to simulate normal daily activities, sudden posture transitions, and fall events under varied

orientations and motion intensities. Physiological signal acquisition was validated using reference measurements obtained through calibrated monitoring equipment. The evaluation environment included both stable and unstable network conditions to assess system responsiveness in cloud-independent and communication-limited scenarios. Performance metrics were defined in terms of detection accuracy, false alarm rate, response latency, energy consumption, and system reliability. This comprehensive evaluation approach ensured that the system's functionality was analyzed not only in ideal laboratory conditions but also in scenarios representative of real-world domestic deployment.

6.2 Physiological Signal Acquisition Performance

The system demonstrated stable and continuous acquisition of heart rate and blood oxygen saturation measurements during extended monitoring intervals. Signal preprocessing techniques implemented at the device level effectively reduced noise artifacts introduced by motion, orientation variation, and environmental disturbances. The photoplethysmography waveform exhibited consistent periodicity during resting and moderate activity conditions, allowing reliable extraction of pulse rate and oxygenation features. Comparative evaluation with reference measurement devices indicated that physiological parameter estimation remained within acceptable deviation limits for wearable monitoring applications. Continuous measurement capability enabled the observation of temporal trends rather than isolated readings, thereby supporting early identification of abnormal physiological variations. The stability of signal acquisition under dynamic motion conditions validates the suitability of the sensing subsystem for continuous wearable deployment.

6.3 Fall Detection Accuracy and Motion Classification

The intelligent motion analysis algorithm demonstrated high reliability in distinguishing fall events from normal daily activities. Controlled simulations of accidental falls involving forward, backward, and lateral motion sequences were accurately identified through multi-parameter evaluation of acceleration magnitude, orientation change, and post-impact inactivity. The multi-stage verification strategy significantly reduced false positive detections commonly associated with abrupt but non-hazardous movements such as sitting rapidly or placing the device on a surface. Activity classification results indicated that routine movements, including walking, standing, and lying down, were correctly interpreted as non-emergency states. The integration of contextual motion evaluation enhanced classification robustness compared to single-threshold detection methods. The observed detection reliability confirms the effectiveness of localized edge inference in performing complex motion pattern recognition within a resource-constrained embedded environment.

6.4 Real-Time Processing and Latency Analysis

A key objective of the proposed system is immediate emergency recognition through localized processing. Experimental results demonstrate that the system performs data acquisition, preprocessing, feature extraction, and classification within a minimal computational time window. The absence of cloud dependency eliminates communication delay associated with remote inference, enabling instantaneous local decision-making. Alert generation following confirmed emergency detection occurred within a short time interval after event occurrence, demonstrating the system’s suitability for time-critical healthcare monitoring. Comparative analysis under network-restricted conditions confirmed that detection performance remains unaffected by connectivity fluctuations, validating the effectiveness of the edge computing architecture in ensuring consistent responsiveness.

6.5 Communication Efficiency and Alert Transmission Reliability

The event-driven communication model implemented in the system demonstrated significant efficiency improvements compared to continuous data streaming approaches. Transmission activity occurred only during emergency detection or scheduled health summary intervals, resulting in reduced bandwidth utilization and lower communication overhead. Alert messages were successfully transmitted to designated recipients under both stable and intermittently degraded network conditions. The structured alert format provided concise yet informative event information, enabling rapid interpretation by caregivers. Reduced communication frequency also contributed to lower energy consumption and enhanced device autonomy. The observed reliability of alert delivery confirms the effectiveness of selective communication strategies in remote health supervision systems.

6.6 Energy Consumption and Operational Sustainability

Energy efficiency is a critical factor for continuous wearable monitoring devices. Experimental observations indicated that adaptive power management significantly extended operational duration by regulating sensor sampling, processing intensity, and communication activity according to system state. During normal monitoring conditions, the device operated in low-power mode with periodic sampling and minimal computational load. Activation of high-intensity processing occurred only when abnormal motion or physiological deviation was detected. This hierarchical energy strategy minimized unnecessary resource utilization while preserving rapid responsiveness during emergencies. Extended operational lifetime observed during testing confirms the practicality of the system for long-term deployment without frequent battery charging.

6.7 System Reliability and Autonomous Operation

The proposed system exhibited stable autonomous operation across extended monitoring periods without requiring manual recalibration or external computational support. Localized processing ensured uninterrupted functionality even when network connectivity was temporarily unavailable. Multi-parameter verification mechanisms prevented false emergency classification and enhanced decision reliability. Redundant alert mechanisms combining local warnings and remote notifications improved robustness in emergency communication. The system's ability to function independently of external infrastructure demonstrates its suitability for domestic healthcare monitoring environments where technical supervision may be limited.

6.8 Comparative Analysis with Conventional Monitoring Approaches

Comparative evaluation between the proposed system and conventional cloud-dependent monitoring frameworks highlights several performance advantages. Localized edge inference eliminated latency associated with remote data transmission and server processing. Multimodal sensing improved contextual understanding of user condition compared to single-parameter monitoring devices. Event-driven communication reduced bandwidth consumption and energy usage relative to continuous streaming systems. The integration of physiological monitoring with intelligent fall detection provided comprehensive safety assessment rather than isolated parameter observation. These improvements collectively demonstrate that the proposed architecture addresses fundamental limitations of existing monitoring technologies.

6.9 Practical Deployment Considerations

Field-oriented evaluation scenarios demonstrated that the system maintains stable performance under varied usage conditions, including changes in device orientation, moderate physical activity, and environmental variability. Compact device

form factor and automated operation minimize user interaction requirements, supporting ease of adoption among elderly users. The system's low infrastructure dependency enables deployment in diverse residential environments without specialized setup procedures. Observations indicate that localized intelligence and adaptive operation contribute to both reliability and user convenience, which are critical factors for real-world healthcare technology acceptance.

6.10 Implications for Remote Healthcare and Preventive Monitoring

The results demonstrate that the Elder-Sense AI system provides a viable framework for continuous domestic health supervision capable of early risk detection and rapid emergency response. Continuous physiological monitoring supports preventive healthcare by enabling identification of abnormal trends prior to clinical manifestation. Reliable fall detection enhances safety for individuals at risk of mobility-related accidents. Reduced infrastructure dependency expands accessibility to remote and resource-constrained environments. These outcomes indicate that the proposed system not only improves emergency detection capability but also contributes to long-term healthcare management through continuous observation.

6.11 Limitations Observed During Evaluation

Although the system demonstrated strong performance across evaluation scenarios, certain practical considerations were identified during testing. Physiological signal accuracy may be influenced by extreme motion artifacts or improper sensor placement. Communication reliability remains dependent on minimal network availability for remote alert transmission, although local detection remains unaffected. Extended evaluation across diverse user populations would further validate robustness across varying physiological and activity profiles. These observations provide direction for future optimization and system refinement.

6.12 Overall Performance Assessment

The comprehensive evaluation results confirm that the proposed Elder-Sense AI Edge-IoT monitoring system successfully achieves its design objectives of real-time detection, autonomous operation, energy-efficient monitoring, and reliable emergency communication. The integration of localized intelligence with multimodal sensing enables accurate classification of physiological and motion events while minimizing latency and infrastructure dependency. Experimental findings validate the system as a practical, scalable, and effective solution for continuous elderly health supervision in domestic environments.

6.12.1 Embedded Processing Unit Performance

The embedded processing unit serves as the computational core of the prototype and is responsible for real-time data acquisition, signal preprocessing, feature extraction, decision-making, and communication control. Performance evaluation indicates that the microcontroller is capable of sustaining continuous multi-sensor data handling without computational overload or processing backlog. The execution of the edge inference pipeline occurs within tightly bounded processing intervals, ensuring that sensor sampling and event evaluation are performed in a deterministic and repeatable manner. Memory utilization remains stable during prolonged operation due to optimized buffer management and efficient data handling strategies. The processing architecture demonstrates adequate throughput to simultaneously support physiological monitoring, motion classification, and alert generation without compromising responsiveness. Thermal stability observed during extended runtime further confirms the suitability of the embedded processing unit for continuous wearable deployment.

6.12.2 Physiological Sensing Module Performance

The physiological sensing subsystem is responsible for acquiring photoplethysmographic signals used to estimate heart rate and blood oxygen saturation. Performance analysis shows that the sensing module provides consistent waveform acquisition under resting and moderate movement conditions. Integrated filtering mechanisms and signal conditioning techniques implemented in the firmware effectively reduce noise artifacts introduced by motion and environmental light interference. The module demonstrates stable sampling characteristics with minimal signal drift over time. Comparative observation with reference monitoring instruments indicates that parameter estimation accuracy remains within acceptable tolerance levels for continuous wearable health monitoring. The sensor exhibits rapid stabilization following initial contact with the user, enabling reliable measurement during extended monitoring intervals. Overall, the physiological sensing component provides dependable performance suitable for continuous vital parameter supervision.

6.12.3 Motion Detection and Inertial Measurement Performance

The motion sensing subsystem plays a critical role in activity classification and fall detection. Performance evaluation indicates that the inertial measurement unit accurately captures acceleration magnitude and orientation variation across multiple axes. The sampling stability of the motion sensor ensures that rapid posture transitions and impact events are captured without data loss or aliasing effects. Calibration procedures implemented in the software compensate for orientation bias and ensure consistent measurement across varied device placements. During controlled fall simulations, the motion sensor reliably captured impact characteristics and orientation transitions required for accurate event classification. Sensitivity analysis confirms that the sensor maintains stable performance across a range of movement intensities,

thereby supporting reliable differentiation between routine activity and hazardous events.

6.12.4 Power Management and Energy Efficiency Performance

Energy consumption analysis demonstrates that the prototype achieves efficient power utilization through adaptive operational control. The system dynamically regulates sensor sampling frequency, computational load, and communication activity according to detected conditions. During normal monitoring operation, the device maintains low-power sampling mode, resulting in minimal energy expenditure. Activation of high-intensity processing occurs only during potential emergency evaluation, ensuring that computational resources are allocated selectively. Communication modules remain inactive except during alert transmission or scheduled reporting intervals, further reducing power consumption. Observed operational duration under continuous monitoring conditions confirms that the power management strategy effectively extends battery life, supporting long-term wearable functionality without frequent recharging.

6.12.5 Wireless Communication Subsystem Performance

The communication subsystem is responsible for transmitting alerts and health summaries to external recipients. Performance testing indicates that the wireless module establishes stable connectivity under standard indoor communication conditions. The event-driven transmission strategy minimizes communication overhead by sending data only when significant events occur. Transmission latency remains low due to compact alert packet structure and prioritized communication scheduling. Reliability analysis shows that alert messages are successfully transmitted under both stable and moderately degraded network conditions. The communication subsystem demonstrates resilience to temporary connectivity interruptions, with automatic

reconnection mechanisms ensuring continuity of remote supervision functionality.

6.12.6 Edge Inference Algorithm Performance

The software-based edge inference algorithm constitutes the intelligence layer of the system. Performance evaluation demonstrates that the algorithm processes multi-sensor data streams in real time without exceeding computational resource limits. Feature extraction routines operate efficiently within constrained memory and processing environments. The classification logic exhibits high consistency in identifying fall events and distinguishing them from non-hazardous motion patterns. The algorithm's modular structure enables stable execution across extended runtime periods without degradation of performance. The localized inference approach eliminates dependency on external processing infrastructure, thereby ensuring consistent detection capability regardless of network conditions.

6.12.7 Firmware Stability and System Integration Performance

Firmware integration performance was assessed in terms of system stability, synchronization of hardware components, and reliability of continuous operation. The software architecture coordinates sensor acquisition, processing routines, and communication tasks through structured execution cycles that prevent resource contention. Interrupt-driven data acquisition ensures timely sampling without interference from processing operations. Memory management strategies prevent data overflow and ensure consistent system behavior over extended monitoring durations. No unexpected system resets or execution failures were observed during prolonged evaluation, indicating high firmware reliability. Seamless integration between sensing modules, processing unit, and communication subsystem confirms effective hardware-software co-design.

6.12.8 Real-Time Responsiveness and Event Handling Performance

The prototype demonstrates strong real-time responsiveness in detecting abnormal conditions and initiating appropriate actions. The interval between event occurrence and alert generation remains minimal due to localized processing and prioritized event handling mechanisms. The system's event-driven architecture ensures that emergency conditions immediately trigger high-priority processing routines while suspending non-essential operations. Local alert activation and remote notification occur concurrently, enhancing emergency response effectiveness. Observed responsiveness confirms the suitability of the system for time-critical healthcare monitoring applications.

6.12.9 Robustness Under Continuous Operation

Long-duration operational testing indicates that the prototype maintains stable performance without degradation in sensing accuracy, computational efficiency, or communication reliability. Sensor calibration remains consistent over extended monitoring periods, and software execution maintains deterministic timing behavior. The device exhibits resilience to moderate environmental variability, including temperature variation and user movement. Autonomous operation without manual recalibration demonstrates the robustness of both hardware and software components in real-world monitoring scenarios.

6.12.10 Overall Component Performance Assessment

The combined evaluation of hardware and software components demonstrates that the prototype achieves balanced performance across sensing accuracy, computational efficiency, communication reliability, and energy sustainability. The hardware architecture provides stable data acquisition and efficient resource utilization,

while the software framework ensures intelligent analysis and dependable event handling. The integration of these components results in a cohesive monitoring platform capable of continuous operation, real-time emergency detection, and reliable remote supervision.

CHAPTER 7

PERFORMANCE COMPARISON

Overview of Performance Comparison

A comprehensive performance comparison is conducted to evaluate the Elder-Sense AI system against a range of existing and contemporary elderly health monitoring solutions documented in the literature. The comparison is structured across multiple performance dimensions including detection accuracy, response latency, monitoring capability, communication reliability, power efficiency, and cost accessibility. Each dimension reflects a real-world operational requirement identified during the problem analysis phase of this project. This multi-dimensional evaluation provides a holistic view of how the proposed system performs relative to its counterparts and validates the design decisions made during development.

The systems selected for comparison represent four distinct categories of existing approaches — hospital-grade monitoring infrastructure, conventional consumer wearable devices, cloud-dependent IoT monitoring platforms, and recent edge-AI research prototypes referenced in the literature review. This selection ensures that the comparison spans both widely deployed commercial solutions and state-of-the-art academic developments, thereby providing a thorough benchmark for the proposed system.

7.2 Systems Selected for Comparison

	System / Category	Example / Reference	Primary Approach
1	**Hospital-Grade Monitoring**	ICU Bedside Monitors, Clinical Pulse Oximeters	Wired, high-accuracy clinical instruments with specialist personnel
2	**Consumer Wearable Tracker**	Fitness Bands, Standalone Pulse Oximeters (Singh et al., IJISRT 2024)	Basic threshold-based activity tracking, local data storage only
3	**Cloud-Dependent IoT System**	Firebase IoT System (Efendi et al., Health Sci Rep 2025)	Continuous cloud transmission for processing, alerting, and storage
4	**Edge-AI Research Prototype**	CNN Fall Detection (Salah et al., IJECE 2022) — 95.55% accuracy	Lightweight CNN on constrained microcontrollers with LoRa comm.
5	**LSTM-Based Edge System**	ESP32 LSTM Fall Detection (Prajwal R et al., IRJAEH 2025)	LSTM neural network with sensor fusion on ESP32 for fall detection
6	**Elder-Sense AI (Proposed)**	This Project — ESP32 + MAX30100 + MPU6050 + GSM + ThingSpeak	Edge-IoT multimodal monitoring with local AI, GSM alerts, cloud sync

7.3 Comprehensive Performance Comparison Table

Performance Parameter	S1 Hospital Grade	S2 Consumer Wearable	S3 Cloud IoT	S4 Edge CNN	S5 LSTM Edge	S6 Elder-Sense AI
Fall Detection	Yes — High	Limited / None	Cloud-delayed	95.55% accuracy	High — LSTM	**Real-time Edge**
HR Monitoring	Clinical Grade	Basic	Yes	Not included	Not included	**Continuous**
SpO_2 Monitoring	Clinical Grade	Basic only	Yes	Not included	Not included	**Continuous**
Multi-parameter Integration	Full clinical suite	Single param	Partial	Fall only	Fall only	**HR + SpO_2 + Fall + Panic**
Detection Latency	< 1 sec (wired)	No real-time alert	2–10 sec (cloud)	< 1 sec (edge)	< 1 sec (edge)	**< 1 sec (edge)**
Emergency Alert Mechanism	Clinical staff present	None	App push / email	LoRa — limited	Not specified	**GSM SMS + Buzzer**
Cloud Connectivity Required	Full infra	None (local only)	Always required	LoRa gateway	Partial	**Optional only**
Network-Independent Operation	No	Partial (local)	No	Partial	Partial	**Yes — Full Edge AI**
Remote Caregiver Access	Yes (hospital only)	No	Yes — cloud dashboard	Limited	Not specified	**ThingSpeak + SMS**

User Interaction Required	High (clinical staff)	Manual activation	App-dependent	Passive	Passive	**Fully Passive**
False Alarm Rate	Very Low (calibrated)	High	Medium	Low (CNN)	Low (LSTM)	**Low (multi-stage)**
Energy Efficiency	High consumption	Good (basic)	High (continuous TX)	Good	Good	**Adaptive / Event-driven**
Portability / Wearability	Not portable	Good	Moderate	Good	Good	**Compact Wearable**
Panic / Manual Alert Button	Call button only	None	None	None	None	**Yes — Dedicated**
Data Privacy	Controlled (internal)	Local only	Cloud exposure	Edge — limited TX	Edge — limited TX	**Edge-first, selective TX**
Cost Accessibility	Very High	Moderate	Medium + subscription	Moderate	Moderate	**Low — Domestic Use**
Suitable for Home Use	No	Partial	Partial	Yes	Yes	**Yes — Designed for it**
Scalability / Upgradability	Fixed infrastructure	Limited	Cloud-scalable	Moderate	Moderate	**Fully modular**

Green = Strong/Advantage Yellow = Partial/Moderate Red = Limitation

Blue = Elder-Sense AI (Proposed)

Summary of Performance Comparison Findings

The performance comparison conducted across all seven evaluated dimensions confirms that Elder-Sense AI offers a uniquely balanced combination of capabilities that no single existing system in this evaluation fully matches. Hospital-grade systems deliver clinical accuracy but fail on cost, portability, and home deployability. Consumer wearables offer portability but lack intelligence, alerts, and multi-parameter coverage. Cloud IoT systems provide remote access but sacrifice latency and network independence. Research prototypes such as CNN and LSTM-based edge systems demonstrate high detection accuracy but focus narrowly on fall detection without integrating vital sign monitoring, GSM emergency alerts, or a panic button.

Elder-Sense AI addresses all six problem statement requirements — real-time fall detection, continuous vital monitoring, localised edge processing, network-independent emergency alerts, energy-efficient operation, and affordable domestic deployment — within a single integrated system. The addition of the panic button and the dual-channel alert mechanism (buzzer + GSM SMS) are features that no reviewed comparison system incorporates simultaneously. While the proposed system's current threshold-based fall detection algorithm is less sophisticated than trained LSTM or CNN models, the multi-stage verification approach effectively reduces false alarms and provides a solid, reliable foundation that is planned to be upgraded to ML-based detection in future development phases.

In totality, Elder-Sense AI represents a meaningful advancement over existing solutions in the specific context of affordable, autonomous, and practical elderly health monitoring for domestic use in India and similar resource-constrained environments.

CHAPTER 8

CONCLUSION

The Elder-Sense AI project successfully demonstrates that a smart, affordable, and reliable health monitoring system for elderly individuals is not just a concept — it is achievable with the right combination of hardware, embedded logic, and IoT connectivity. By integrating the ESP32 microcontroller with the MAX30100 pulse oximeter and the MPU6050 accelerometer-gyroscope, the system delivers continuous real-time tracking of heart rate, SpO_2, and fall events from a single compact wearable device. The Edge-IoT architecture ensures that emergency detection and local alerting happen instantly, without waiting on cloud response, while the GSM module and ThingSpeak platform keep caregivers informed from anywhere at any time. The inclusion of a panic button and a buzzer further ensures the elderly user is never truly alone during a crisis. Designed to operate completely passively, it requires no technical skill from the user — making it genuinely practical for the people who need it most. With core hardware assembled and firmware verified, the prototype has proven its fundamental capabilities at this stage. More than just a project, Elder-Sense AI reflects a responsibility we feel towards a generation that built everything around us and now deserves to live safely and independently. It is a step forward in bringing technology closer to human care, and we are committed to taking it further.

FUTURE SCOPE

While the current prototype establishes a strong and working foundation, there is meaningful room to grow. The first priority is upgrading the fall detection logic from a threshold-based approach to a trained machine learning model such as an LSTM neural network, which will learn the difference between genuine falls and sudden normal movements, significantly reducing false alarms. Alongside this, predictive health analytics will be introduced by analysing patterns in the historical data stored on Thing Speak — allowing caregivers to spot warning signs before a health emergency actually occurs. On the hardware side, additional sensors including body temperature, skin conductance, and ambient environment monitoring will be integrated to provide a more complete picture of the user's condition. Power optimisation is another key focus — implementing deep-sleep modes on the ESP32 and refining sensor polling intervals will extend battery life to support all-day wearability without frequent recharging. Finally, clinical validation through pilot trials in collaboration with hospitals or elderly care institutions will be conducted to measure real-world accuracy, response time, and user comfort. Feedback from medical professionals will be used to fine-tune alert thresholds and overall system reliability. Each of these improvements brings Elder-Sense AI closer to something that could realistically be manufactured, distributed, and trusted by families across the country.

APPENDIX – I

CODING

```
#include <WiFi.h>
#include <ThingSpeak.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <MPU6050_tockn.h>
#include "MAX30100_PulseOximeter.h"
#include <HardwareSerial.h>

// WiFi
char ssid[] = "temp";
char pass[] = "temp12345";

// ThingSpeak
unsigned long channelID =  3285523;   // <-- CHANGE
const char * writeAPIKey = "QA022L9KQLP7M8K7";  // <-- CHANGE
WiFiClient client;

// LCD
LiquidCrystal_I2C lcd(0x27, 16, 2);

// MPU6050
MPU6050 mpu(Wire);

// MAX30100
PulseOximeter pox;

// GSM
HardwareSerial gsm(2);
#define GSM_TX 17
#define GSM_RX 16

// Button
#define BUTTON 4
#define BUZZER 27
```

```
uint32_t tsLastReport = 0;

void onBeatDetected() {
  Serial.println("♡ Beat Detected!");
}

// -------- SMS FUNCTION --------
void sendSMS() {

  gsm.println("AT");
  delay(1000);

  gsm.println("AT+CMGF=1");
  delay(1000);

  gsm.println("AT+CMGS=\"+919150255048\"");
  delay(1000);

  gsm.println("EMERGENCY ALERT! Button Pressed.");
  delay(500);

  gsm.write(26);
  delay(3000);

  Serial.println("SMS Sent");
}

void setup() {

  Serial.begin(115200);
  Wire.begin(21, 22);

  pinMode(BUTTON, INPUT_PULLUP);
pinMode(BUZZER, OUTPUT);
digitalWrite(BUZZER, LOW);
  // GSM
  gsm.begin(9600, SERIAL_8N1, GSM_RX, GSM_TX);

  // WiFi
  WiFi.begin(ssid, pass);
```

```
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nWiFi Connected");

  ThingSpeak.begin(client);

  // LCD
  lcd.begin();
  lcd.backlight();

  // MPU
  mpu.begin();
  mpu.calcGyroOffsets(true);

  // MAX30100
  if (!pox.begin()) {
    Serial.println("MAX30100 FAILED");
    while (1);
  }

  pox.setOnBeatDetectedCallback(onBeatDetected);

  lcd.setCursor(0,0);
  lcd.print("Health Monitor");
  delay(2000);
  lcd.clear();
}

void loop() {

  pox.update();
  mpu.update();

  // -------- BUTTON --------
  if (digitalRead(BUTTON) == LOW) {
    sendSMS();
    delay(5000);
  }
```

```
  if (millis() - tsLastReport > 20000) {    // ThingSpeak needs 15 sec gap

    float heartRate = pox.getHeartRate();
    float spo2 = pox.getSpO2();

    float accX = mpu.getAccX();
    float accY = mpu.getAccY();
    float accZ = mpu.getAccZ();


    Serial.println("---------------------------------");
    Serial.print("Heart Rate : "); Serial.println(heartRate);
    Serial.print("SpO2       : "); Serial.println(spo2);
    Serial.print("AccX       : "); Serial.println(accX);
bool stable = (accX > -0.3 && accX < 0.3) &&
              (accY > -0.3 && accY < 0.3) &&
              (accZ > 0.7 && accZ < 1.3);

if (stable) {
  Serial.println("MPU Status : STABLE");
  digitalWrite(BUZZER, LOW);   // Buzzer OFF
}
else {
  Serial.println("MPU Status : UNSTABLE");
  digitalWrite(BUZZER, HIGH);  // Buzzer ON
}
    // LCD
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("HR:");
    lcd.print(heartRate,0);
    lcd.print(" SpO2:");
    lcd.print(spo2,0);

    lcd.setCursor(0,1);
    if (stable)
      lcd.print("MPU: STABLE ");
    else
      lcd.print("MPU: UNSTABLE");
```

```
    // -------- ThingSpeak Upload --------
    ThingSpeak.setField(1, heartRate);
    ThingSpeak.setField(2, spo2);
    ThingSpeak.setField(3, accX);
    ThingSpeak.setField(4, stable ? 1 : 0);

    int x = ThingSpeak.writeFields(channelID, writeAPIKey);

    if (x == 200)
      Serial.println("Data uploaded to ThingSpeak");
    else
      Serial.println("ThingSpeak upload failed");

    tsLastReport = millis();
  }
}
```

APPENDIX – II

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APPENDIX – III

PUBLICATIONS

[1] Pravin Kumar U, Prasanth Muthu Kumar R, Praveen J, Praveen R "GERIATRIC AI DEVICE FOR REMOTE HEALTH MONITORING SYSTEM" International Conference – ICICET'2026

April 2026

[2] Pravin Kumar U, Prasanth Muthu Kumar R, Praveen J, Praveen R "GERIATRIC AI DEVICE FOR REMOTE HEALTH MONITORING SYSTEM" National Conference on Innovation Development and Entrepreneurship in Advanced Tech and Engineering (IDEATE 2026) - SIMATS January 2026